PMx Weekly Digest: November 09, 2026

Content Source: Full Text

Paper 1: Assessment of Allometric Scaling Methods for the Prediction of Human Pharmacokin

Bibliographic Information

- **Title**: Assessment of Allometric Scaling Methods for the Prediction of Human Pharmacokinetic of LNA-i-miR-221.
- **Authors**: Hadija Marchiori, Michele Schiavon, Chiara Roversi, Simone Zannoni, Massimiliano Fonsi, Pierfrancesco Tassone, Pierosandro Tagliaferri, Maria Teresa Di Martino, Alessia Tagliavini, Chiara Dalla Man
- **Journal**: CPT: Pharmacometrics & Systems Pharmacology
- **DOI**: 10.1002/psp4.70359

Paper Type: POPK_PKPD

Quick Take

This study systematically evaluates single-species allometric scaling methods to translate the pharmacokinetics of the locked nucleic acid antisense oligonucleotide LNA-i-miR-221 from preclinical species (rats and monkeys) to humans. The authors demonstrate that simple allometric scaling with an empirical exponent ($\beta = 0.85$) accurately predicts human clearance (fold-error 1.05), whereas volume of distribution predictions carry substantial uncertainty (within 2- to 4-fold error). It is essential reading for translational pharmacometricians designing first-in-human trials for oligonucleotide therapeutics seeking to balance 3Rs animal reduction with safe starting dose selection.

Executive Summary

Translating preclinical pharmacokinetics to first-in-human (FIH) trials for antisense oligonucleotides (ASOs) remains a major translational hurdle due to complex distribution kinetics, high tissue uptake, and non-canonical clearance pathways. Marchiori and colleagues retrospectively assessed various single-species allometric scaling algorithms using preclinical data from 42 Sprague Dawley rats and non-human primate data to predict human PK parameters of LNA-i-miR-221 (a 2′–4′ methylene-bridged chemically modified ASO targeting miR-221), which were validated against a 15-patient Phase I dose-escalation oncology trial. Both rat and human PK were characterized using a three-compartment model. Simple allometry with fixed allometric exponents outperformed correction factors (brain weight, maximum lifespan potential, and plasma protein binding); simple allometry with $\beta = 0.85$ predicted human systemic clearance (CL) with high fidelity (fold error: 1.05 ± 0.30, $p = 0.27$). In contrast, distribution volumes were less reliably translated, with only peripheral compartment volume V_2 yielding fold-error near unity ($\beta = 0.8$), while central (V_1), deep peripheral (V_3), and steady-state distribution volume (V_d) exhibited up to 4-fold errors. Simulated human profiles reproduced observed C_{max} across several cohorts but underestimated AUC at the highest dose levels, likely reflecting uncaptured dose-

dependent tissue saturation. These findings provide a pragmatic single-species scaling framework for ASOs while underscoring that wider safety margins are mandatory when relying on single-species volume extrapolations.

Scientific Context & Motivation

MicroRNAs such as miR-221 frequently function as oncogenes in hematological and solid malignancies, making them high-priority targets for RNA-targeted therapeutics. Locked nucleic acid (LNA) antisense oligonucleotides incorporate a 2′-O,4′-C-methylene bicyclic ribose modification that dramatically elevates hybrid duplex thermal stability and in vivo nuclease resistance. Despite favorable pharmacology, translational allometric scaling of ASOs from preclinical species to humans is notoriously fraught with inconsistency. Most canonical allometric scaling paradigms (e.g., Dedrick plots, physiological time, rule-of-exponent approaches) were established for small-molecule xenobiotics eliminated primarily by cytochrome P450 enzymes or glomerular filtration. ASOs undergo minimal hepatic CYP metabolism and display high intracellular uptake (predominantly in liver and kidney), plasma protein binding to albumin, and endonuclease/exonuclease cleavage. Furthermore, standard multi-species allometry requires at least three species (rodents, dogs, and non-human primates), contradicting modern 3Rs (Replacement, Reduction, Refinement) regulatory initiatives. Although isolated studies have explored ASO scaling, consensus on robust single-species scaling algorithms—particularly when parameterized within structural population compartmental PK frameworks—has been lacking. This study directly addresses this knowledge gap by benchmarking multiple literature-derived single-species scaling equations for LNA-i-miR-221 against empirical Phase I clinical data.

⚡ Methodological Snapshot

The authors leveraged preclinical PK data from 42 Sprague Dawley rats receiving intravenous bolus doses of LNA-i-miR-221 across two 4-day cycles separated by a 10-day washout period (dose range: 5 to 125 mg/kg), and clinical PK data from a Phase I dose-escalation trial involving 15 cancer patients divided across five intravenous infusion cohorts. Both preclinical and clinical concentration-time profiles were characterized using an identical three-compartment structural model parameterized with systemic clearance (CL), central distribution volume (V_1), peripheral volumes (V_2, V_3), inter-compartmental distribution clearances (BCL_2, BCL_3), and an initial lag time (T_{lag}) in rats.

Single-species allometric scaling was conducted using individual empirical Bayes estimates from rats. Candidate scaling formulations included: (1) simple allometry ($Y = \alpha \cdot W^{\beta}$) with fixed literature exponents ($\beta = 0.75, 0.8, 0.85, 1.0$ for CL; $\beta = 0.8, 0.95, 1.0, 1.1$ for volumes); (2) averaged multi-exponent allometry; (3) correction factors incorporating brain weight (BrW), maximum life span potential (MLP), and unbound fraction (f_u); (4) the empirical Tang method; and (5) a weight-independent model. Scaled human parameters were evaluated against observed human population parameters using fold-error metrics and Wilcoxon signed-rank tests. Finally, forward deterministic simulations using the best-performing scaled parameters were evaluated against median human concentration-time profiles across cohorts to assess peak concentrations (C_{max}),

half-life, and AUC accuracy, alongside an exploratory cross-species comparison utilizing non-human primate non-compartmental analysis (NCA) data.

⌧ Structural Model Breakdown

A three-compartment model with linear elimination from the central compartment and delayed dose appearance in rats described both species:

1. Central Compartment (Q_1, Volume V_1): $\frac{dQ_1(t)}{dt} = -(k_{01} + k_{21} + k_{31}) \cdot Q_1(t) + k_{12} \cdot Q_2(t) + k_{13} \cdot Q_3(t) + D \cdot \delta(t - T_{\text{lag}})$
2. First Peripheral Compartment (Q_2, Volume V_2): $\frac{dQ_2(t)}{dt} = -k_{12} \cdot Q_2(t) + k_{21} \cdot Q_1(t)$
3. Second Peripheral Compartment (Q_3, Volume V_3): $\frac{dQ_3(t)}{dt} = -k_{13} \cdot Q_3(t) + k_{31} \cdot Q_1(t)$
4. Observation Equation: $y(t) = \frac{Q_1(t)}{V_1}$

Clearance reparameterizations were defined as: $CL = k_{01} \cdot V_1$ $BCL_2 = k_{21} \cdot V_1 = k_{12} \cdot V_2$ $BCL_3 = k_{31} \cdot V_1 = k_{13} \cdot V_3$ $V_d = V_1 + V_2 + V_3$

Rat population parameter estimates (Table 1): $T_{lag} = 0.04$ h (RSE 22.2%), $CL = 0.11$ L/h (7.26%), $V_1 = 0.04$ L (8.60%), Q_2 (BCL_2) = 0.22 L/h (15.1%), $V_2 = 0.15$ L (8.99%), Q_3 (BCL_3) = 0.03 L/h (8.59%), $V_3 = 1.85$ L (27.3%). Human population parameter estimates (Table 2): $CL = 200$ mL/h/kg (RSE 8.09%), $V_1 = 45$ mL/kg (58.2%), $Q_2 = 480$ mL/h/kg (27.3%), $V_2 = 200$ mL/kg (16.4%), $Q_3 = 68$ mL/h/kg (22.7%), $V_3 = 930$ mL/kg (16.3%).

Detailed Methodological Analysis

Modeling Approach

Nonlinear mixed-effects (NLME) modeling was utilized to generate population and empirical Bayes individual PK parameter estimates in rats and humans using an identical 3-compartment linear structural model. Single-species allometric translation was conducted by applying empirical power laws and correction algorithms directly to the individual rat parameter distributions to project predicted human parameter distributions.

Data Sources

Preclinical data: 42 Sprague Dawley rats from a dose-range finding study (5 to 125 mg/kg IV bolus administered daily across two 4-day cycles with 10-day washout). Clinical data: Phase I dose-escalation clinical study (EudraCT 2017-002615-33, NCT04811898) consisting of 15 adult cancer patients across 5 cohorts receiving intravenous infusions of LNA-i-miR-221 (dense sampling during and post-infusion on cycle 1, sparse sampling across washouts). Monkey data for exploratory comparison were extracted from historical NCA literature (single-dose IV administration).

Estimation Methods

Population PK estimation parameters were obtained using maximum likelihood algorithms (FOCE-I / stochastic estimation as documented in the primary model development citations).

Individual empirical Bayes post-hoc parameter distributions were scaled using point-by-point transformation to preserve preclinical biological variability.

Model Evaluation

Prediction accuracy was primarily quantified by fold-error ($r = Y_{pred}/Y_{obs}$) and acceptance within the standard 2-fold window ($0.5 \leq r \leq 2.0$). Statistical departure from ideal unity ($r = 1.0$) was evaluated using the Wilcoxon signed-rank test across the individual predicted distributions. Deterministic profile simulations with scaled human parameters were evaluated against clinical cohort medians via percentage estimation error for C_{max}, terminal half-life, and AUC.

Covariate Analysis

Preclinical-to-clinical scaling used body weight (W) as the primary physiological scaling covariate. Evaluated biological correction factors included brain weight (BrW), maximum life span potential (MLP), and in vitro plasma unbound fraction (f_u).

Statistical Rigor Assessment

The study applies appropriate non-parametric statistical testing (Wilcoxon signed-rank test) to evaluate whether fold-error distributions deviate significantly from unity. Rather than relying solely on population point estimates, rat individual empirical Bayes parameter estimates were scaled to generate realistic distributions of predicted human parameters. However, statistical power in the clinical validation dataset is inherently limited by the small sample size of the Phase I trial (15 patients total, 3 per cohort), precluding meaningful between-subject variability characterization in humans. Additionally, the cross-species comparison with monkeys is confounded because monkey data were derived via non-compartmental analysis from a single animal dataset, whereas rat data were estimated via a population mixed-effects compartmental model.

☒ Key Findings

1. Simple Allometry Outperforms Complex Corrections for Clearance: Simple allometry with an exponent of $\beta = 0.85$ provided the most accurate prediction of human clearance (predicted CL: 14.67 ± 4.28 L/h vs. observed: ~ 14.0 L/h; fold error: 1.05 ± 0.30; $p = 0.27$, not significantly different from 1). Standard allometric exponents ($\beta = 0.75$, $\beta = 0.80$) yielded underpredictions (fold errors of 0.60 and 0.80, respectively), while $\beta = 1.0$ severely overpredicted CL (fold error: 2.46). Correction factors (BrW, MLP) failed catastrophically (fold errors 0.001 to 0.03).

2. Distribution Volumes Exhibit Substantial Prediction Uncertainty: Single-species scaling for volumes of distribution proved challenging. While V_2 was accurately scaled with $\beta = 0.8$ (fold error: 1.06 ± 0.28, $p = 0.09$), central volume V_1 was overpredicted (fold error: 1.45, $p < 0.001$), and deep peripheral volume V_3 as well as total volume V_d had large prediction errors (fold errors of 2.62 to 3.01 for $\beta = 0.8$; and > 8.0 for $\beta = 1.0$), falling outside the acceptable 2-fold margin and extending into a 4-fold error window.

3. Concentration Profile Simulations: Applying the best-performing scaled parameters (CL $\beta = 0.85$; volumes $\beta = 0.8$) in the 3-compartment model reproduced human C_{max} accurately at high doses (estimation errors -2% in Cohort 1, $+7\%$ in Cohort 4, -7% in Cohort 5), but overpredicted peaks by $\sim 44\%$ in intermediate cohorts. Predicted systemic exposure (AUC) was accurate at low doses (Cohort 1 error: $+2\%$) but significantly underpredicted observed AUC at the highest clinical dose (Cohort 5 error: -35%), pointing toward saturable clearance or non-linear accumulation in humans.
4. Rodent vs. Non-Human Primate Scaling: When comparing NCA-derived monkey scaling to rat scaling, monkeys yielded superior total distribution volume predictions (V_d fold error: 1.01 with $\beta = 0.8$), whereas rat popPK scaling yielded slightly superior clearance predictions (CL fold error: 1.05 vs. monkey CL fold error: 0.90 with $\beta = 1.0$).

Clinical & Regulatory Implications

For clinical pharmacologists and translational teams developing oligonucleotide therapeutics, these results demonstrate that single-species rat popPK data can reliably inform human clearance projections using simple allometry with $\beta = 0.85$, supporting mechanistic starting-dose calculations for FIH trials without requiring extensive non-human primate usage. However, because volume of distribution parameters—specifically peripheral tissue distribution (V_3)—were poorly predicted (up to 3- to 4-fold error), initial C_{max} and prolonged tissue accumulation cannot be anticipated with high precision. Consequently, starting doses calculated from rat allometric models must incorporate conservative safety margins (e.g., lower starting dose tiers, widened dose escalation steps, and conservative exposure stopping rules). Furthermore, the observation of underpredicted AUC at high clinical doses suggests saturable distribution or elimination processes, emphasizing the necessity of real-time PK monitoring during clinical dose escalation.

Strengths & Limitations

Strengths

- Head-to-head empirical validation of multiple literature allometric scaling algorithms against true human Phase I clinical PK data for a locked nucleic acid antisense oligonucleotide.
- Use of identical three-compartment structural population PK models in both preclinical (rat) and clinical (human) populations, eliminating structural model discordance.
- Propagation of preclinical between-animal variability through empirical Bayes estimates to generate statistical fold-error distributions evaluated via non-parametric hypothesis testing.
- Direct evaluation of full concentration-time profiles and exposure metrics (C_{max}, AUC, half-life) rather than restricting analysis to isolated scalar clearance and volume parameters.

Limitations (Acknowledged by Authors)

- Validation was restricted to a small clinical cohort of 15 oncology patients ($n = 3$ per cohort), limiting statistical power to dissect inter-individual variability.

- Preclinical rat studies exhibited potential bioanalytical assay insensitivity at lower dose levels, complicating definitive assessment of dose linearity.
- The monkey comparison relied on non-compartmental analysis from a single historical dataset, confounding species differences with analytical methodology.
- Findings are compound-specific to LNA-i-miR-221 and cannot be assumed a priori to generalize to ASOs with different sequences, lengths, backbone chemistries, or GalNAc conjugations.

Limitations (Expert Review)

- The authors did not evaluate physiologically based pharmacokinetic (PBPK) modeling or quantitative systems pharmacology (QSP) frameworks, which can explicitly account for endocytic cell uptake and tissue phosphorothioate/LNA binding mechanisms.
- Inter-compartmental clearances (Q_2/BCL_2 and Q_3/BCL_3) were not explicitly detailed in the allometric scaling comparisons, despite having a major influence on simulated distribution phases and terminal half-life predictions.
- Non-linear kinetics observed in clinical AUC at higher dose levels were modeled as strictly linear, potentially skewing exposure predictions at Phase II therapeutic doses.

Generalizability

The specific allometric exponent identified for clearance ($\beta = 0.85$) and volume ($\beta = 0.80$) is directly applicable to unconjugated LNA oligonucleotides administered intravenously. However, generalizability to GalNAc-conjugated ASOs, morpholinos, or siRNA therapeutics is restricted, as receptor-mediated liver uptake and lipid nanoparticle formulations dramatically alter biodistribution kinetics away from classical allometric relationships.

Key Equations

Three-Compartment Systemic PK Differential Equations

$$\begin{cases} \frac{dQ_1}{dt} = -(k_{01} + k_{21} + k_{31}) \cdot Q_1(t) + k_{12} \cdot Q_2(t) + k_{13} \cdot Q_3(t) + D \cdot \delta\big(t - T_{\text{lag}}\big), & Q_1(0) = 0 \\ \frac{dQ_2}{dt} = -k_{12} \cdot Q_2(t) + k_{21} \cdot Q_1(t), & Q_2(0) = 0 \\ \frac{dQ_3}{dt} = -k_{13} \cdot Q_3(t) + k_{31} \cdot Q_1(t), & Q_3(0) = 0 \\ y(t) = \frac{Q_1(t)}{V_1} \end{cases}$$

Three-compartment structural pharmacokinetic model with linear elimination from the central compartment and delayed absorption/dose appearance (T_{lag}) used to describe LNA-i-miR-221 disposition in rats and humans (Equation 1).

Standard Simple Allometric Power Law

$$Y = \alpha \cdot W^{\beta}$$

Power-law function relating a pharmacokinetic parameter Y (clearance or volume of distribution) to body weight W using allometric coefficient α and fixed scaling exponent β (Equation 5).

Average Multi-Exponent Allometric Scaling

$$Y = \frac{1}{3}\sum_{i=1}^{3} \alpha_i \cdot W^{\beta_i}$$

Averaged allometric projection equation combining scaling predictions across three distinct literature exponents (Equation 6).

Single-Species Rat Clearance Scaling (Tang Method)

$$CL_{\text{human}} = 0.152 \cdot CL_{\text{rat}} \cdot \frac{W_{\text{human}}}{W_{\text{rat}}}$$

Empirical single-species rat-to-human clearance extrapolation formula specifically proposed in literature for oligonucleotide therapeutics (Equation 10).

Weight-Independent Clearance Scaling (Mahmood Method)

$$CL_{\text{human}} = 33.35 \cdot \left(\frac{\frac{a}{f_{u,\ \text{rat}}}}{f_{u,\ \text{hum}}} \right)^{0.77}$$

Weight-independent allometric scaling equation incorporating plasma protein binding ratio correction for clearance prediction (Equation 11).

Allometric Prediction Fold-Error

$$r = \frac{Y_{\text{pred}}}{Y_{\text{obs}}}$$

Metric evaluating the accuracy of the allometrically predicted human parameter relative to the true estimated human population parameter (Equation 12).

Figures & Tables

- **Figure 1**: Observed median human plasma concentration-time profiles (red circles/dashed lines) versus simulated profiles using scaled rat PK parameters (black lines) across all five Phase I dose cohorts.
  - *Significance*: Illustrates that rat-scaled allometric parameters capture peak concentration timing and initial rapid elimination, but underestimate terminal persistence and exposure at higher dose levels.
- **Table 1**: Population pharmacokinetic parameter estimates and relative standard errors (RSE%) for the three-compartment model in Sprague Dawley rats.
  - *Significance*: Provides the baseline preclinical parameter estimates ($CL = 0.11$ L/h, $V_1 = 0.04$ L, $V_2 = 0.15$ L, $V_3 = 1.85$ L) utilized for allometric scaling.
- **Table 2**: Population pharmacokinetic parameter estimates and relative standard errors (RSE%) for the three-compartment model in human cancer patients.
  - *Significance*: Establishes the ground-truth clinical parameters ($CL = 200$ mL/h/kg, $V_1 = 45$ mL/kg, $V_2 = 200$ mL/kg, $V_3 = 930$ mL/kg) against which scaling methods were benchmarked.

- **Table 3**: Comprehensive comparison of rat-to-human allometric scaling algorithms for CL, V_1, V_2, V_3, and V_d, showing predicted values, fold-error (mean $\pm$ SD), and Wilcoxon test p-values.
  - *Significance*: Key result table establishing simple allometry with $\beta = 0.85$ as optimal for CL (fold-error 1.05) and demonstrating that correction factors (BrW, MLP) fail dramatically for ASOs.
- **Table 4**: Comparison of observed versus simulated C_{max} across four daily peaks in Cohorts 1 through 5.
  - *Significance*: Demonstrates that peak concentrations were accurately projected in Cohorts 1, 4, and 5 (errors within $\pm 7\%$), but overestimated by $\sim 44\%$ in intermediate cohorts.
- **Table 5**: Comparison of observed versus simulated half-life across daily peaks in Cohorts 1 through 5.
  - *Significance*: Quantifies a modest systematic underestimation of human distribution/elimination half-life ($\sim$ 21 minutes shorter on average) by the scaled model.
- **Table 6**: Comparison of observed versus simulated AUC across Cohorts 1 through 5.
  - *Significance*: Highlights non-proportionality at high doses, where simulated AUC matched Cohort 1 within 2% but underestimated Cohort 5 AUC by 35%.
- **Table 7**: Allometric scaling results derived from non-human primate (monkey) NCA data for human CL and V_d.
  - *Significance*: Demonstrates that monkey data yield an exceptionally accurate steady-state volume prediction (V_d fold error = 1.01 with $\beta = 0.8$) while rat data provided slightly better CL predictions.

Supplementary Materials

The supplementary document (psp470359-sup-0001-supinfo.docx) contains comprehensive details regarding the chemical structure, molecular weight, modified locked nucleic acid backbone configuration (2′-O,4′-C-methylene bridge), and bioanalytical hybridization assay validation for LNA-i-miR-221.

Future Directions

1. Multi-Species Integration: Conduct prospective cross-species comparisons where both rodent and non-human primate datasets are modeled using identical structural popPK/PBPK methodologies rather than contrasting compartmental with non-compartmental analysis.
2. Mechanistic PBPK and QSP Modeling: Develop physiologically based models incorporating saturable endocytosis, tissue uptake transporters, and oligonucleotide plasma protein binding to capture the non-linear AUC accumulation observed at elevated human doses.
3. Sequence and Chemistry Cross-Validation: Systematically apply this allometric benchmarking workflow to other chemical classes of ASOs (e.g., 2′-MOE gapmers, phosphorothioate backbones, GalNAc conjugates) to establish whether $\beta = 0.85$ is an invariant property of oligonucleotide renal/endonuclease clearance or unique to short LNA designs.

Expert Commentary

This paper tackles an urgent, highly practical problem in model-informed drug development (MIDD): how to safely and rationally set first-in-human starting doses for oligonucleotide therapeutics while adhering to 3Rs principles. Pharmacometricians frequently encounter regulatory skepticism when using single-species rodent allometry for large molecules or non-CYP cleared agents. Marchiori et al. provide compelling evidence that simple allometry for systemic clearance works remarkably well when parameterized with an exponent of $\beta = 0.85$ (fold error: 1.05), substantially outperforming the traditional small-molecule exponent of 0.75 (which underpredicts CL by 40%). Crucially, the paper serves as a strong cautionary tale against blind application of small-molecule physiological correction factors: incorporating brain weight or maximum lifespan potential resulted in 30- to 1000-fold errors, because ASO elimination is not governed by hepatic metabolic turnover or oxidative stress longevity biology.

However, the paper reveals an equally important vulnerability: volume of distribution scaling remains tenuous. Extrapolating deep peripheral compartment volume (V_3) and total V_d resulted in 3- to 4-fold errors. For antisense drugs, tissue uptake into the liver, renal cortex, and spleen is driven by high-affinity endocytic processes and intracellular protein binding. A simple body-weight power law cannot resolve differences in tissue receptor density or organ perfusion across species. The authors' simulation exercise in Figure 1 also illustrates that linear scaled models break down at higher dose levels (35% AUC underprediction in Cohort 5), reflecting either bioanalytical limits or saturable tissue sequestration. As a teaching point for translational pharmacometricians: while single-species rat allometry with $\beta = 0.85$ can comfortably anchor clearance for human dose predictions, volume uncertainty and non-linear tissue uptake mandate conservative safety margins and intensive early-cohort pharmacokinetic monitoring during FIH dose escalation.

Bottom Line

Single-species allometric scaling from rat population PK models using simple allometry with an exponent of $\beta = 0.85$ accurately predicts human systemic clearance of LNA-i-miR-221 within a 1.05-fold error, while standard correction factors fail. However, distribution volumes cannot be scaled with equal confidence (2- to 4-fold error), meaning FIH starting dose designs utilizing single-species rodent data must incorporate robust safety margins to mitigate risks associated with tissue accumulation and non-linear exposure.

Fact-check corrections

Figures

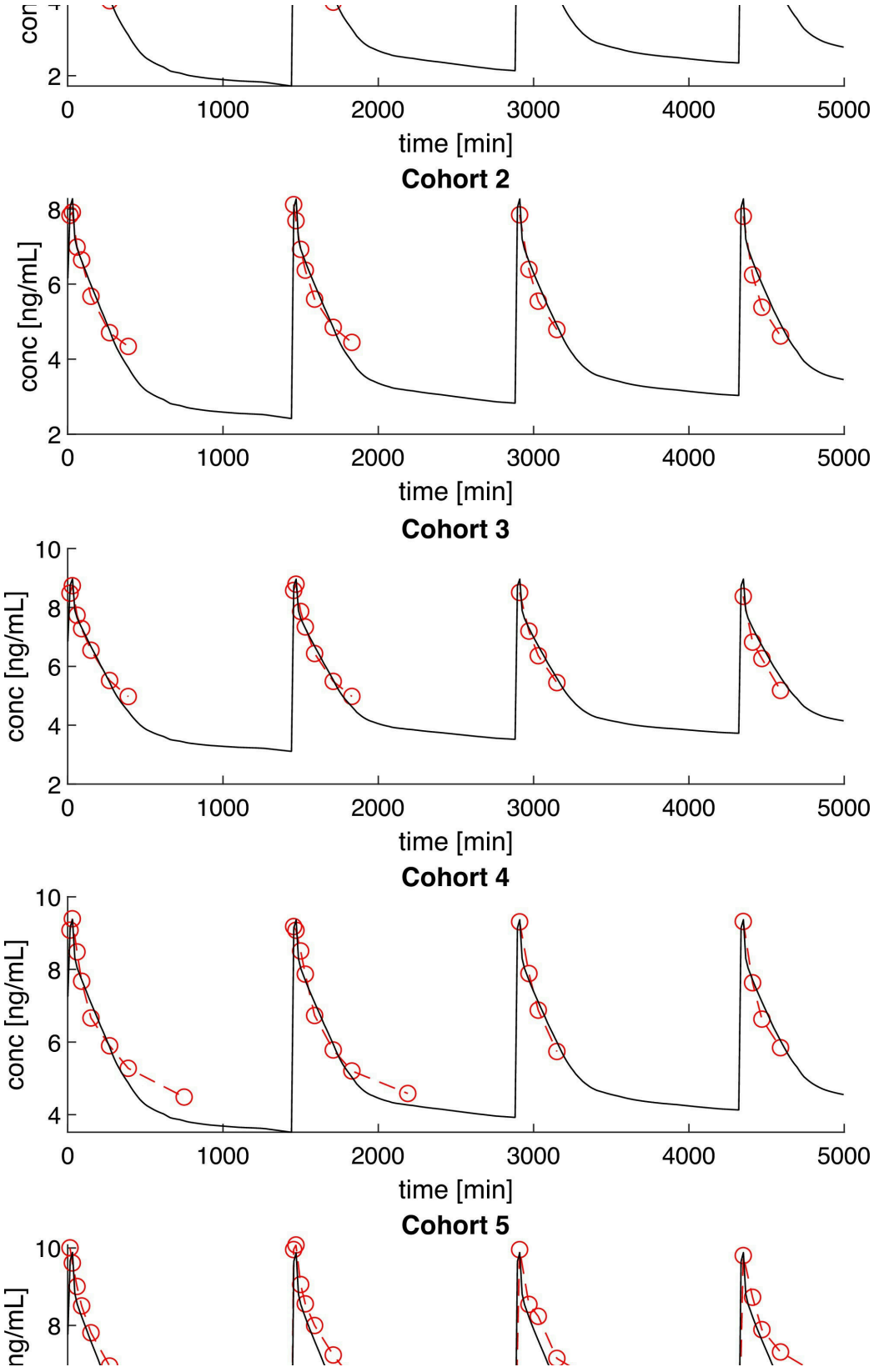


Figure 1: Expected concentration of LNA-i-miR-221 in humans (black line) vs median concentration measured during a FIH clinical trial (dotted red line with circular markers). The p